is that it will happen. And if something can happen in *enormously* more ways, like the pages landing in the wrong numerical order, it is *enormously* more likely that it will happen. We all know this intuitively. If you buy one lottery ticket, there is only one way you can win. If you buy a million tickets, each with different numbers, there are a million ways you can win, so your chances of striking it rich are a million times higher.

Entropy is a concept that makes this idea precise by counting the number of ways, consistent with the laws of physics, in which any given physical situation can be realized. *High entropy means that there are many ways; low entropy means there are few ways.* If the pages of *War and Peace* are stacked in proper numerical order, that is a low-entropy configuration, because there is one and only one ordering that meets the criterion. If the pages are out of numerical order, that is a high-entropy situation, because a little calculation shows that there are
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